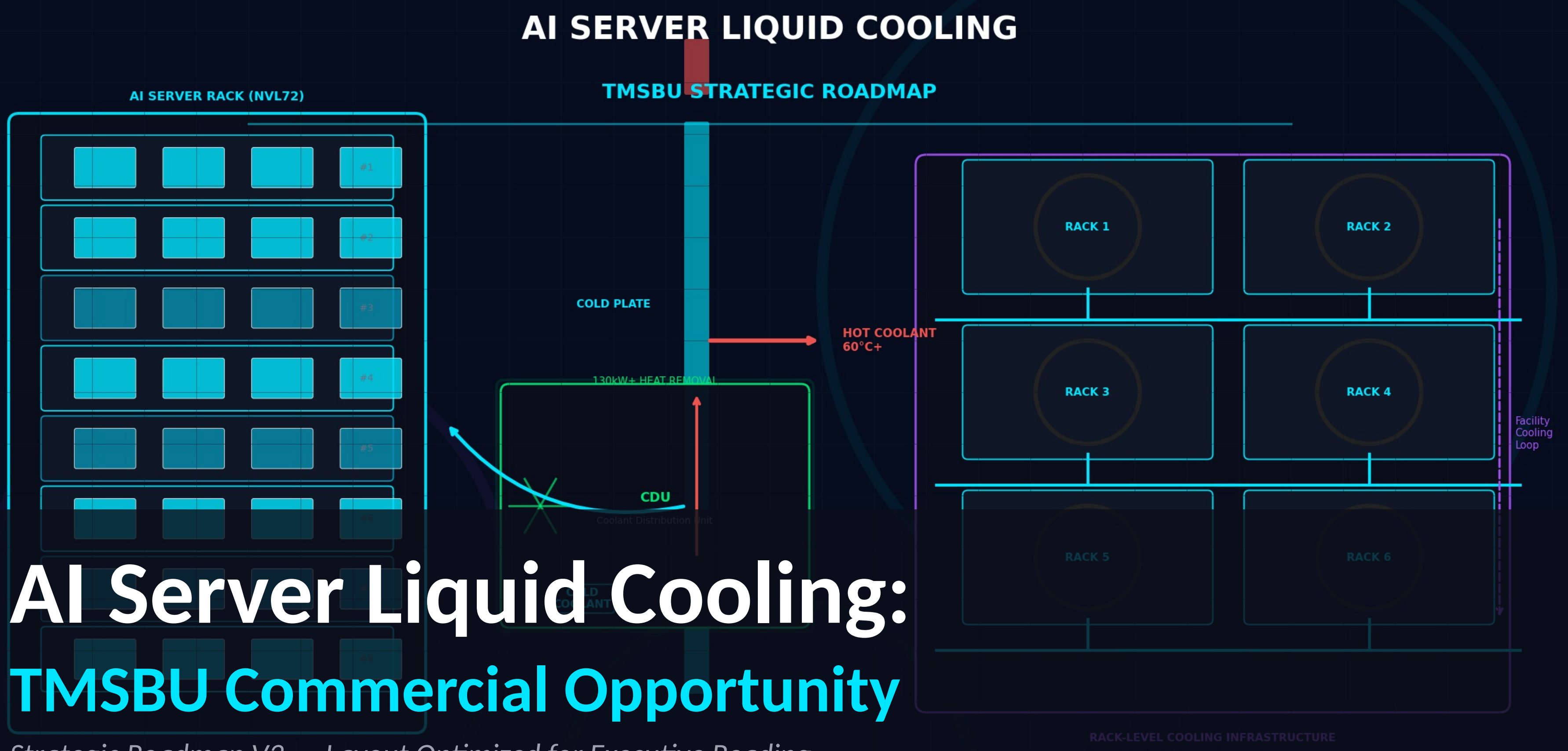


AI Server Liquid Cooling:

TMSBU Commercial Opportunity

Strategic Roadmap V3 — Layout Optimized for Executive Reading

Executive Summary

液冷是商業模式轉型，不是產品升級

TREND 01

1,400W GPU TDP by 2027+

GPU Power Exceeds Air Cooling

GPU TDP by 2027+

MARKET 02

60%+ new AI DC by 2026

Liquid Cooling Is Standard

new AI DC by 2026

HYBRID 03

BOTH ，TMSBU 兩者皆受益

Air + Liquid Coexist

，TMSBU 兩者皆受益

OPPORTUNITY 04

$1.8B SOM addressable (assumptions apply)

TMSBU: Component → System Solution

SOM addressable (assumptions apply)

ACTION 05

NOW Phase 1 需立即啟動

6–18 Months: Critical Window

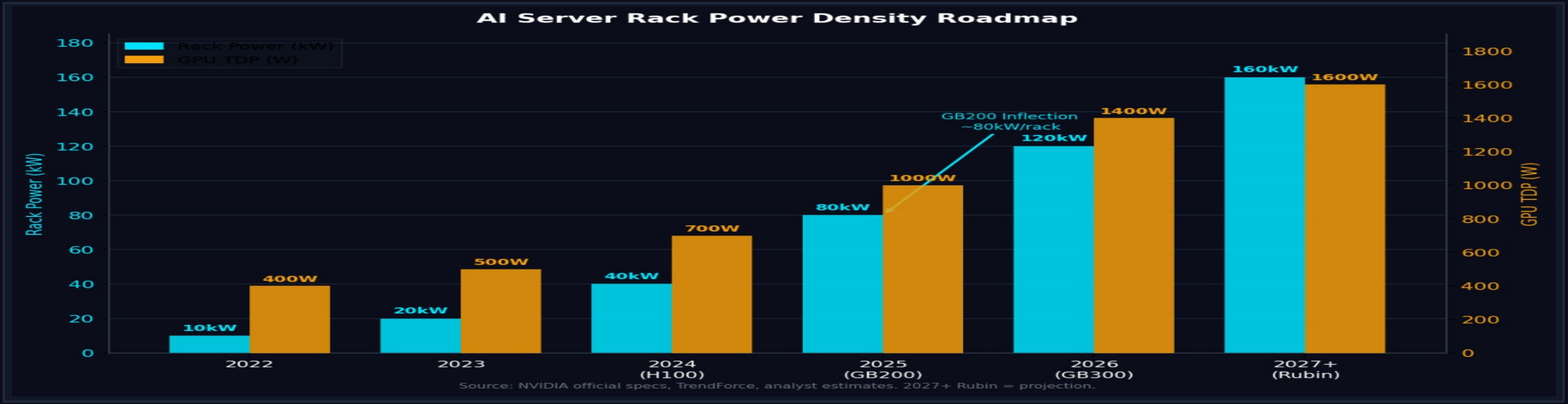
Phase 1 需立即啟動

💡 **The future belongs to thermal system integrators — not component suppliers.**

Sources: MarketsandMarkets (TAM) | Gartner (adoption rate) | NVIDIA official specs | TMSBU analysis

Market Inflection: AI Compute Density Creates Thermal Bottleneck

GPU TDP 突破風冷散熱上限，散熱瓶頸成為 AI 基礎建設擴張的硬約束



1,400W
GPU TDP @ 2027+ Rubin

160kW
Rack Power @ 2027+ (est.)

$39.1B
Liquid Cooling TAM by 2030

KEY INSIGHTS

- GB200 NVL72 （ ~80–130kW/rack ）是風冷物理瓶頸的轉折點
- Rubin （ 2027+ ）功率密度 160kW+ ，浸沒式冷卻可能成下一個轉折
- 散熱瓶頸 = 誰能解決散熱、誰就能贏得 AI 基礎建設市佔

Sources: NVIDIA official GB200 specs | TrendForce rack power analysis | MarketsandMarkets TAM | Rubin = AnandTech projection

Why Air Cooling Alone Is No Longer Enough

Not Air vs. Liquid — The Future Is Hybrid Architecture

LIQUID COOLING

Handles GPU / CPU Hotspot

- GPU TDP 700W–1,400W+ 直接導熱
- Cold Plate 直接接觸 die，效率 > 95%
- CDU 130kW+ 熱移除能力
- Manifold 分流至多 server

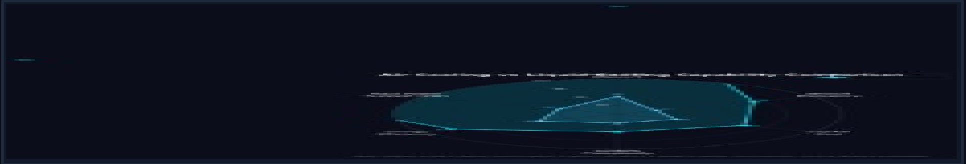
+

AIR COOLING

Handles Memory / PCB / Power

- HBM Memory（每模組 ~1kW）散熱
- PCB / 電源供應模組
- Networking / Storage 元件
- 低功率 accessory 元件

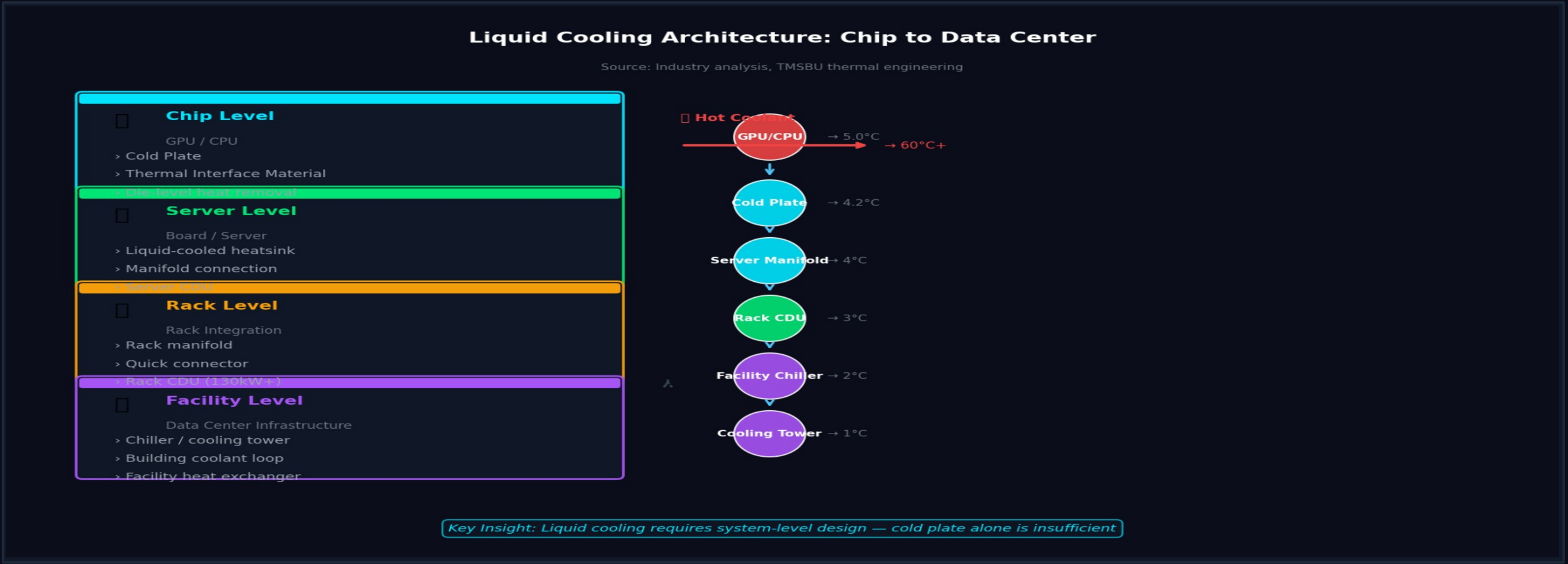
Not Air vs. Liquid — The Future Is Hybrid Architecture



TMSBU BENEFIT: Fan / VC / HS demand ↑ with liquid cooling adoption (more non-GPU components need air cooling)

Liquid Cooling Architecture Becomes the New Standard

液冷的價值在完整系統整合： Cold Plate → CDU → Manifold → Quick Connector



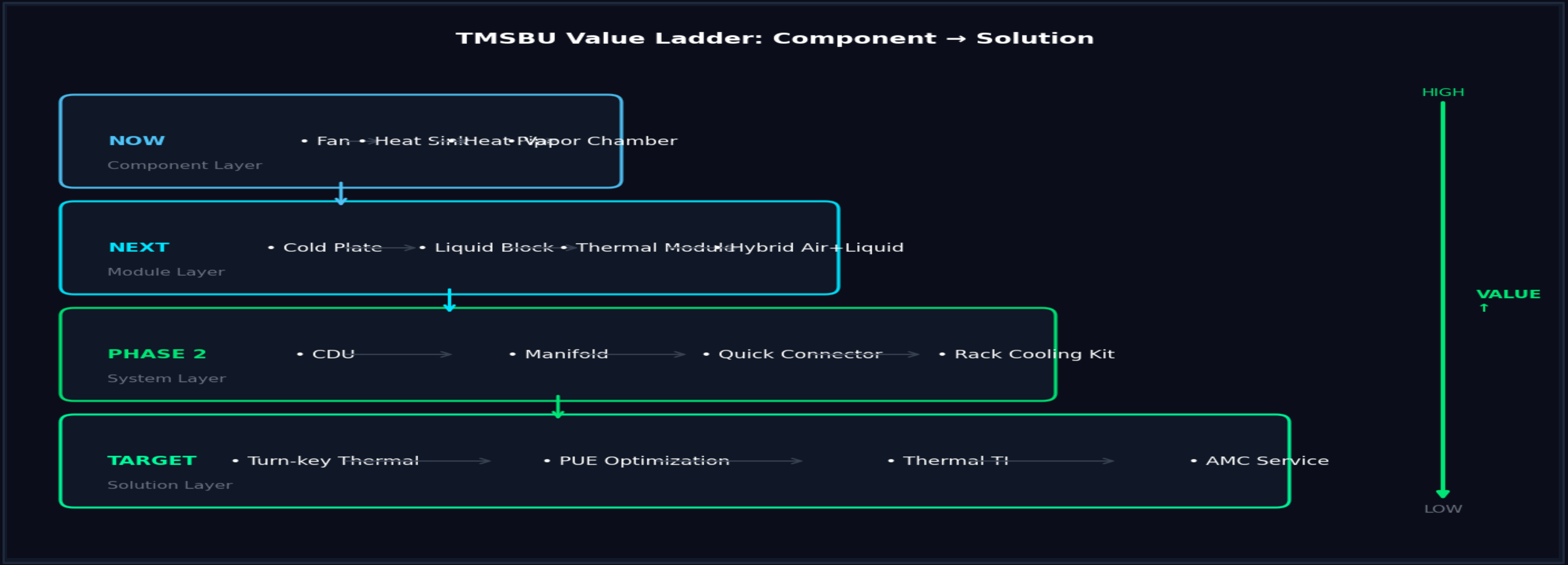
KEY INSIGHTS

- Cold Plate + CDU + Manifold + Quick Connector + Leak Detection = 完整液冷系統
- 系統整合複雜度是最高附加價值的來源 — 不是單一零件
- Vendor 能提供完整系統者，才能拿下最高市佔與毛利

Source: Industry analysis, TMSBU thermal engineering. Components: CoolIT, Rittal, Staubli public specs.

TMSBU Portfolio Mapping:
From Component to System Solution

自然擴張路徑：零件 → 模組 → 系統 → 解決方案



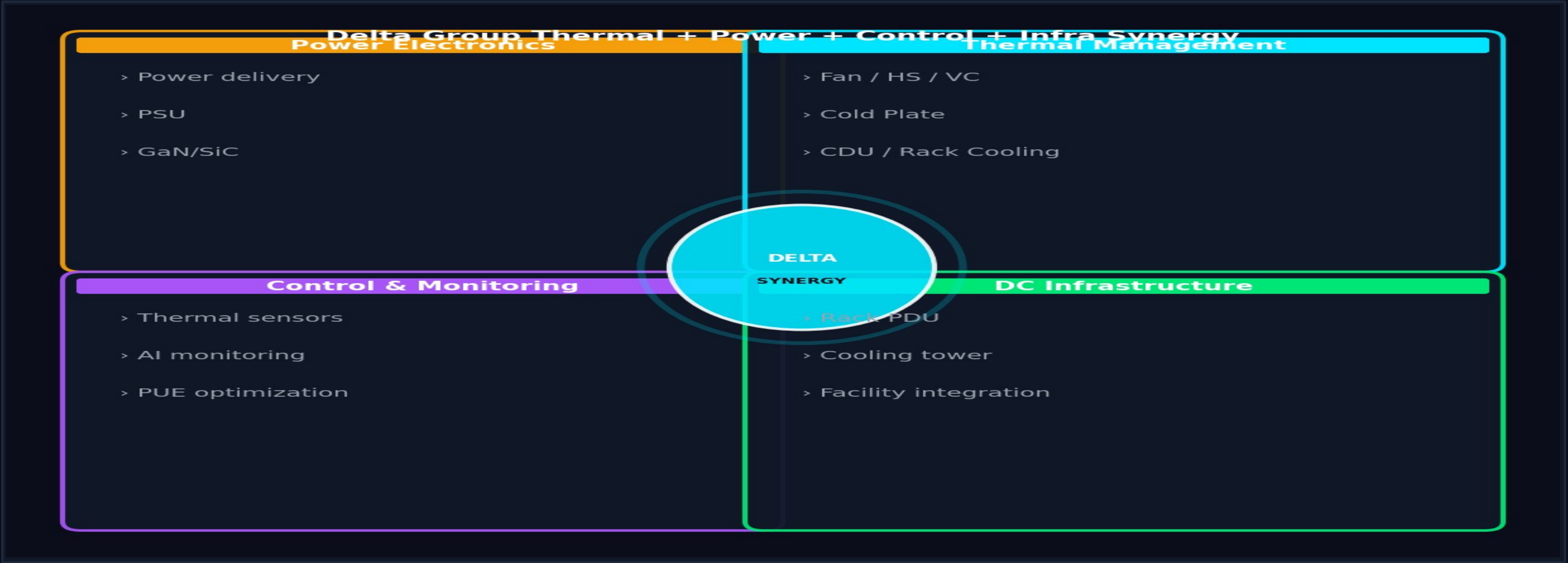
KEY INSIGHTS

- CDU + Quick Connector + Manifold = TMSBU 最高價值的擴張方向
- 從零件報價升級到系統方案統包報價，改變商業模式
- 目標：成為客戶眼中不可或缺的 Thermal System Partner

Source: TMSBU internal product portfolio analysis

Delta Group Synergy:
Why TMSBU Can Win in Liquid Cooling

唯一同時具備風冷與液冷能力的供應商 + Delta 集團層級綜效



KEY INSIGHTS

- Delta 集團唯一能 Chip-level → Rack-level → Facility-level 全段熱管理整合
- Fan + Power + Control + DC Infra 集團綜效為競爭對手無法快速複製的護城河
- Manufacturing scale + Global support network 支撐 Tier-1 客戶信任

Source: Delta Electronics public information, TMSBU analysis

Market Opportunity: TAM / SAM / SOM

系統層級解決方案擴大 TMSBU 可觸及市場規模

KEY INSIGHTS

- TAM $39.1B（M&M 2024）涵蓋整個 AI 資料中心液冷市場
- SOM = 假設 TMSBU 滲透率 14% of SAM，需要 James 內部驗證
- 3 情境：保守 $0.8B / 基準 $1.8B / 積極 $3.2B — 取決於 TMSBU 執行速度

⚠ SOM figures are assumptions — need internal TMSBU validation. TAM: MarketsandMarkets. SAM: analyst estimate (~33% of TAM). Scenarios: TMSBU analysis.

Capability Gap & Risk Assessment

3 項最高優先風險： CDU 能力 / 洩漏可靠性 / 客戶認證週期

Risk Item	Impact	Urgency	Priority Action
CDU Design & Manufacturing	HIGH	HIGH	核心系統零件。合作或收購， Phase 1 決策
Leak Detection & Reliability	HIGH	HIGH	單次洩漏可摧毀 $50 萬 server。可靠性 non-negotiable
Customer Qualification Cycle	MED	HIGH	12-18 個月 design-in cycles。立即啟動
Quick Connector Qualification	HIGH	MED	Staubli/Liiver 夥伴關係確認
Coolant Compatibility	MED	MED	OEM 專屬 coolant specs ，需早期 engagement
Thermal Simulation & Testing	MED	MED	Chip-to-rack 多物理場模擬能力建立
Integration Responsibility	HIGH	MED	系統銷售邊界與保固責任需提前定義
Global Support Network	MED	LOW	Tier-1 hyperscaler 合約門檻
China Supply Chain Price War	MED	MED	冷板 / Manifold 成本競爭，以品質差異化
Talent / Skill Gap	MED	MED	液冷系統工程師人才儲備

IMMEDIATE PRIORITIES

① CDU Strategy (in-house / partner / acquire) | ② Leak reliability (quick connector qualification) | ③ Customer qualification cycle (start NOW)

Source: TMSBU internal risk assessment. Urgency = time sensitivity. Impact = business consequence if risk materializes.

Strategic Roadmap & Recommended Actions

4 階段 6-18 個月 Roadmap ：Phase 1 基礎 → Phase 4 規模化

Phase 1: 0–3M

onfirm CDU partner • Benchmark GB200/GB300 • uild product gap list • Identify target customers

Phase 2: 3–6M

Cold plate + fan module proto • Validation equipment • Thermal TI discussions • Business case

Phase 3: 6–12M

CDU/manifold kit proto • Quick connector qualification • Rack cooling demo • Tier-1 engagement

Phase 4: 12–18M

Solution selling model • AMC service design • Reference customer • Mass production plan

KEY INSIGHTS

- Phase 1 最重要決策：CDU 是內部開發 / 合作夥伴 / 併購？這將決定 TMSBU 液冷策略成敗
- 所有 Phase 的最終目標：從 Component Supplier 轉型為 Thermal System Solution Provider
- 落後的代價：失去 GB300/Rubin 平台的供應商資格，影響未來 5-10 年競爭地位

Source: TMSBU strategic planning analysis. Timelines are indicative and subject to internal alignment.